GIDFINE LOAN PLATFORM

Final Pre-Push Release Audit Report — Strict Read-Only Mode

Branch: development	**Starting HEAD:** 931411f	**Main Hash:** f7ed8d1 (Untouched)
Ahead of Remote: 5 commits	**Frontend Suites:** 13/13 (52/52 Pass)	**Python Regressions:** 6/6 Pass

1. Git Integrity & Working Tree Status

- **HEAD Commit:** 931411f exactly 5 commits ahead of origin/development.
- **Staged Changes:** 0 files (working tree cleanly staged/unstaged for explicit review).
- **Tracked Modifications:** 4 files (schema.ts, api.d.ts, AdminDashboard.tsx, loanService.ts).
- **Untracked Implementations:** ledger.ts, ledgerTypes.ts, CashierLedgerPortal.tsx, financialLedgerWorkflow.test.ts, ledgerSchema.test.ts.

2. Financial Engine Implementation Guarantees

- **Integer Pesewas:** All calculations strictly integer pesewas; negative numbers and decimals rejected.
- **Double-Entry Balance:** validateDoubleEntryBalance() verifies Total Debits == Total Credits == Total Transaction Amount.
- **Maker-Checker Enforcement:** Cryptographically verified via ctx.auth.getUserIdentity(). Maker cannot approve own request.
- **Waterfall Repayments:** Verified order: 1. Fees first, 2. Interest second, 3. Principal remainder.
- **Compensating Reversals:** Original transactions remain immutable; dual-authorization writes balanced inverted entries.

3. Concurrency & Optimistic Concurrency Control (OCC)

- Convex runtime (v1.47.0) executes mutations under serializable ACID transactions with automatic conflict retry.
- Indexed queries (by_agency_and_idempotency) ensure concurrent requests for identical keys serialize and return identical records.
- Repayments re-read unpaid balances dynamically to prevent schedule double-deduction under concurrent submission.

4. Test Reproducibility Summary

Test Suite Name	Command	Status
Financial Ledger Workflow	node --experimental-strip-types --test tests/financialLedgerWorkflow.test.ts	8/8 Pass
Ledger Schema & Idempotency	node --experimental-strip-types --test tests/ledgerSchema.test.ts	12/12 Pass
Combined Frontend Suites	node --experimental-strip-types --test tests/*.test.ts	52/52 Pass (13 Suites)
Python FastAPI Engine	pytest tests/ -q	6/6 Pass
TypeScript Compiler Validation	npx tsc --noEmit	0 Errors (Exit 0)

5. Unresolved Items for Matthias & Final Commit Plan

- **Pending Decisions:** Administrator reversal bypass exception, repayment overpayment liability policy, and delinquency restructuring rules (all non-blocking for baseline MVP).
- **Commit Plan:** Single atomic commit ready for push: feat(ledger): implement Phase 4 financial engine, double-entry journal, maker-checker disbursements, and cashier UI.
- **Audit Confirmation:** Strictly read-only; no files modified, no commits made, no remote pushes executed.